

Exact and Structural Methods for the Job Sequencing and Tool Switching Problem

State of the work

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<https://github.com/shankar-n/JGP-SSP-Codes-Shankar>

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Outline

- 1 The problem and the two questions
- 2 A worst-case theory of the heuristic
- 3 Which structure decides the gap?
- 4 Two exact solvers
- 5 The computational study
- 6 Remaining work

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Why this problem

Flexible manufacturing. A CNC machine serves many part types from a tool magazine of capacity b ; every tool switch halts production while the magazine is reconfigured — switches are *the* controllable cost [Crama et al. 1994].

Beyond machining. The same structure — a capacitated store serving set-valued, reorderable requests — appears in PCB assembly, warehouse picking, and colour printing (ink cartridges as tools) [Burger et al. 2015].

Two live gaps in the literature. The heuristic everyone uses has no worst-case theory; the exact state of the art solves nothing at $n \geq 30$. This project attacks both — and they meet at one number, $|U| - b$.

The problem

Data. Jobs J , tools T , magazine capacity b ; job j requires $T_j \subseteq T$, $|T_j| \leq b$.

Decision. An order of the jobs and a loading policy; each tool insertion costs 1.

- Fixed order $\Rightarrow$ loading solved exactly by KTNS (keep tool needed soonest) in $O(n|T|)$ [Tang–Denardo 1988]
- Joint problem NP-hard [Crama et al. 1994]

Practice: group, then sequence. Partition jobs into fewest magazine-fitting groups (JGP), order the groups (small TSP), run KTNS. Applied verbatim in industry; assessed only *empirically* until now [Burger et al. 2015].

Two questions, one quantity

Q1 (heuristic)

How far can group-then-sequence be from optimal? When is it exact?

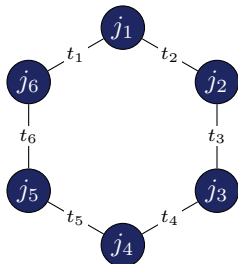
Q2 (exact methods)

Can the SSP be solved through tool *configurations* (column generation, Benders), as the JGP so successfully is?

The organising quantity: the coverage bound $|U| - b$

Every used tool beyond the initial load must be inserted once, so $Z^* \geq |U| - b$. It is the LP value of every natural configuration relaxation (proved) — and on part of the instance space it *is* the optimum.

The smallest counterexample: the 6-ring



$$T_j = \{j, j+1 \bmod 6\}, b = 3$$

- optimum: $Z^* = 3$ — a *four-group* walk, one tool per transition
- heuristic ($K^* = 3$ groups, best order):
 $H = 4$

gap = 1, ratio = $4/3$

The grouping phase is to blame: minimal cardinality hides the cheaper grouping. g disjoint copies $\Rightarrow$ gap = g : **unbounded**.

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The gap identified exactly

Theorem (Grouping exactness)

$Z^* = \min_{\mathcal{P}} \gamma(\mathcal{P})$ over all feasible groupings. Hence $\text{gap} = H - Z^* =$ the price of restricting to minimum cardinality.

Proved zero-gap classes (every b): $K^* \leq 2, \quad |U| \leq b + 1, \quad Z^* \leq 2.$

The setup-cost spectrum: SSP $\longleftrightarrow$ JGP

In practice a changeover costs a fixed setup ρ beyond the per-tool swaps.

Theorem (Both ends explicit)

$\rho < \frac{1}{n-1} \Rightarrow$ the augmented problem is the SSP.

$\rho > H - Z^* \Rightarrow$ every optimum uses minimum-cardinality grouping: the heuristic is exactly optimal.

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- the transition lives on $[1/(n-1), H - Z^*]$: it is governed by the *gap itself*;
- real tooling reports $\rho \approx 3\text{--}60 \Rightarrow$ production sits far on the collapsed (JGP) side — a rigorous licence for standard practice.

The worst case, quantitatively

For $K^* = 3$ (configurations $\subseteq U$, $q = |U| - b$, $R = Z^* - q$):

$$H_{\text{walk}} = q + \min \min(x_{ab}, x_{bc}, x_{ca}), \quad x_{uv} = |(C_u \cap C_v) \setminus C_w|$$

$$\text{gap} \leq \max\left(0, \min\left(q, \left\lfloor \frac{2b-q}{3} \right\rfloor\right) - R\right) \quad (\text{walk value; tight at every known walk ext})$$

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$$\frac{H}{Z^*} \leq \min(b, K^* - 1, q)$$

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Zero-gap classes, every b : $K^* \leq 2$, $|U| \leq b + 1$, $Z^* \leq 2$; and
 $Z^* \leq 3 \Rightarrow \text{gap} \leq 1$.

Determining the worst case: a certificate, and a separation

Natural candidate (all data $b \leq 4$): $\text{gap} \leq K^* - 2$. Two readings of the heuristic: hold each group's configuration (*walk* value H_{walk}), or re-evaluate the flattened order with KTNS ($H \leq H_{\text{walk}}$, the method as defined).

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Certificate ($b = 5$, $K^* = 3$, $|U| = 9$)

$T = (\{0, 1, 2, 4, 7\}, \{0, 4, 5\}, \{1, 5\}, \{2, 6, 8\}, \{3, 4, 5, 8\})$: $Z^* = 4$, $H_{\text{walk}} = 6$ — the candidate **fails for the walk model**, exactly on the boundary of the proved regimes, attaining our bound with equality.

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The separation: on this very witness KTNS beats the walk: $H = 5$, true $\text{gap} = 1$ — and every known walk-extremal instance collapses the same way. The candidate is *reopened* for the heuristic itself; new: $Z^* \leq 3 \Rightarrow \text{gap} \leq 1$ (every b , every K^*).

The $4/3$ ratio, and what remains

Theorem ($b = 3$)

Every instance with $K^ \leq 3$ satisfies $H/Z^* \leq 4/3$.*

- ingredients: gap ≤ 1 at $K^* = 3, b = 3$ (counting law) and positive gap $\Rightarrow Z^* \geq 3$ (small-optimum lemma);
- the case $K^* \geq 4$ is the *one* surviving conjecture — populated (witnesses with ratio $5/4$ exist), no violation known;
- the edge-family pattern “positive gap $\Rightarrow K^* = 3$ ” does *not* extend — found and corrected this month.

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Which structure decides these questions?

Negative (gap side). Two instances with the *same* circuit clutter, same K^* , same Z^* have different gaps (0 vs 1) $\Rightarrow$ no job-compatibility invariant can decide exactness; tool-level overlaps are essential.

Theorem (Covering side: odd rings, via blocking duality)

Clusters over maximal groups; blocker = minimal covers (K^ = its min size). On the k -ring, $b = 3$: fractional cover = $k/2$, $K^* = \lceil k/2 \rceil$ — **tight iff k even.***

Odd circular structure: first provable obstruction for the set-covering relaxation behind JGP branch-and-price — the parallel of odd holes.

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Solver 1: branch-and-Benders-cut

Why Benders fits. Fixed order $\Rightarrow$ loading is polynomial (KTNS). Put the *order* in an integer master, settle the *loading* exactly in a subproblem:

$$\min \theta \quad \text{s.t.} \quad \text{Ham. path via depot,} \quad \theta \geq \sum_{ij} w_{ij} x_{ij}, \quad \theta \geq |U|, \quad \text{cuts}$$

- loading subproblem = totally unimodular LP = KTNS $\Rightarrow$ *exact* Benders optimality cuts from its dual;
- plus combinatorial cuts and triplet bounds; subtours separated exactly at integer candidates; 2^3 cut configurations ablated;
- the coverage row $\theta \geq |U|$ is our theory's one-line fix — the re-run measures exactly its effect.

Solver 2: position formulations and branch-and-price

The obstacle. The natural configuration model needs exponentially many subtour constraints; MTZ repairs are exact only with one potential per configuration — exponential again (proved).

The route: fix a polynomial row set, $O(n|T|)$, generate columns.

- **PCF'**: column = configuration at a position; rows: one configuration per position, cover every job, count insertions. LP value = $|U| - b$ (proved) — the diagnostic instrument;
- **PTF**: column = *transition* ($C \rightarrow C'$) per step, with tool-by-tool flow consistency; $LP \geq |U| - b$ and *strictly above* on witnesses — first in the family to beat the bound;
- MILP pricers; robust branching on aggregated $a_{t,p}$ keeps the pricing problem structurally intact.

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The computational instrument

Why run it. The theorems single out $|U| - b$; the benchmark measures it across the standard suites. And a prior question: the SSP's polynomial loading subproblem is the textbook precondition for Benders — untested until now.

- methods: 8 Benders configs (ours), LSS '04, Catanzaro-F4 '15, SSPMF '24, our branch-and-price (PCF', PTF)
- 1,421 standard instances ($n \leq 40$, $|T| \leq 60$); 3600 s / 600 s; pinned threads; early stop after 8 straight failures
- one canonical metric: every returned sequence re-scored by KTNS
- integrity: 806 instances solved by someone — **zero** cross-method disagreements
- code, baselines, and per-instance results:

<https://github.com/shankar-n/JGP-SSP-Codes-Shankar>

Experimental design

Environment. LIMOS SLURM cluster, homogeneous AMD EPYC 7452 nodes; CPLEX 22.1.1, 4 dedicated threads per run (B&P single-threaded, PySCIPOpt); 16 GB/run; time limits 3600 s (primary) / 600 s (secondary); all other solver parameters at defaults.

Protocol. Instances ordered easiest-first by $(n, m, b, \text{density})$, identically for every method; a configuration is retired after 8 consecutive non-optimal runs. Attempted counts are therefore *outcomes*; results are reported over uniform suite denominators.

Validation. All exact solvers pre-checked against brute force on small instances; every returned sequence re-scored with KTNS in a single cost convention; 806 instances solved by ≥ 1 method, **zero** cross-method disagreements.

Results

solved (uniform denominators)	primary, of 411	secondary, of 1,010
SSPMF (2024)	138	(not run, by design)
LSS (2004, modern CPLEX)	91	667
Catanzaro F4 (2015)	91	455
Benders, corrected master (ours)	43	244 (best arm 274)
PCF' B&P (ours) [†]	1	93
PTF B&P (ours) [†]	1	71

Unattempted counts as unsolved — every method over the same denominator. [†]prototype scope $n \leq 25$.

Per-family solved/attempted detail: report, Table 2.

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Per-family solved/attempted detail: report, Table 2.

- SSPMF strongest on primary — the published SOTA claim *replicates*
- 2004 formulation + modern solver $\approx$ 2015 formulation, and dominates secondary: solver generation outweighs formulation vintage
- corrected Benders, where it solves: geomean 1.0s on the 4-method common set vs 333/616s for the '04/'15 compact models
- everyone's wall: $n = 15$ (at $|T| \approx 20$); nobody solves $n \geq 30$

The diagnosis: bound-limited, mechanically

Pre-fix: optimum found in seconds, *proof* bound frozen for the full hour ($\sim 250k$ nodes, $\sim 75k$ cuts); fractional cuts **structurally inactive**. **With the coverage row** $\theta \geq |U|$: $1/9 \rightarrow 43/73$ (primary), $26/48 \rightarrow 244/318$ (secondary); 122/287 solves at the root node, median 0.05 s; the ablation flattens; *all* 104 remaining timeouts still hold the known optimum — nothing outside the bound-tight class is solved.

The measured split

Among the 806 solved instances: 48.0% have $Z^* = |U| - b$ exactly (upper-biased estimate). The suites split in half between the regime where heuristic and exact methods win at the root, and the regime where new mathematics is required.

PCF'/PTF confirm the same law from the branch-and-price side (15% of PCF' optima at the root; PTF's stronger bound is the one lever that exceeds $|U| - b$).

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- ❶ **Implemented, cluster re-run queued:** the Benders master with the coverage row $\theta \geq |U|$ — verified; on bound-tight instances the proof now closes *at the root, with zero cuts* (the 6-ring: 0 cuts vs. 443 before).
- ❷ **Theory (open):** the exact worst case of the KTNS-evaluated heuristic (is the counting bound the truth?); the 4/3 case $K^* \geq 4$.
- ❸ **With Prof. Wagler:** PTF polytope facets on the known witnesses (PORTA); the odd-ring idealness question for JGP.

Questions for today: which side of the blocking pair (clusters/covers vs. circuits/groups) to pursue first; priority between the Benders re-run and the polytope study; and whether the remaining theory items match your expectations for the closing period.